

Meter/CT Self-Check



Background

Meter and CT is the Measuring device installed at the GCP (Grid Connection Point) of the inverter system for detecting GCP data. For energy storage system, the functions such as Battery Charge and Discharge, Working Mode, Export Control and Three-phase Unbalanced, can only be achieved when the Measurement device is installed correctly.

The Meter/CT Self-Check function can identify whether there has problems with the Meter and CT installation through the feedback of the power changes. After the installation is completed, you just need to use this function, then you can troubleshoot the problems on site.

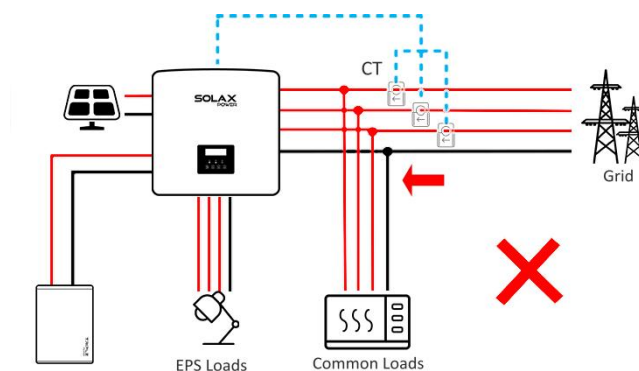
What problems can be troubleshooted with Meter/CT Self-Check

Function?

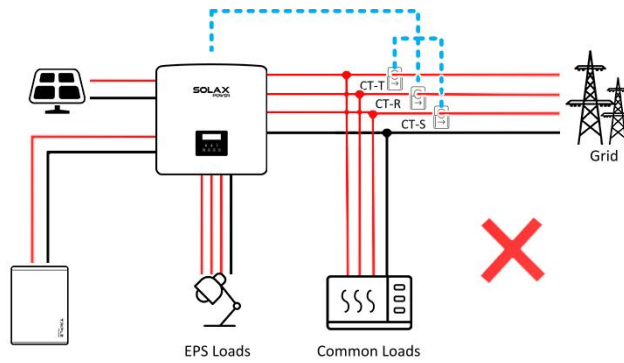
Some problems can be corrected automatically, and others need to be manually adjusted by the installer based on the error code displayed after the checking is completed.

(1) For the following Installation problems, the inverter will adjust the correspondence with Measurement Device to **Automatically Correct** them:

① 'CT' direction is reversed



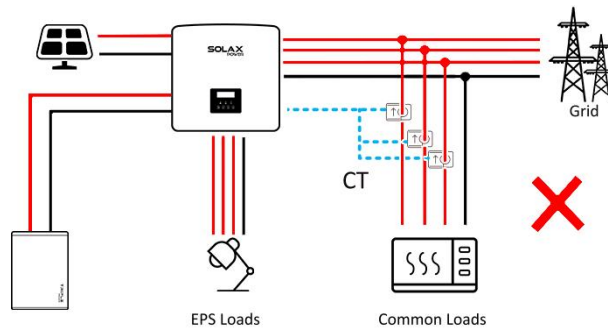
② 'CT' three phases correspondence is not correct



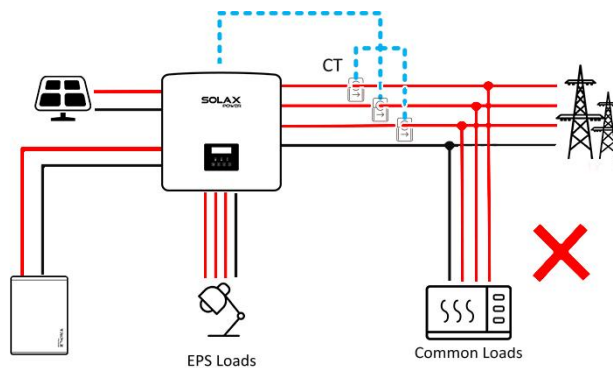
Note: Meter's three phases correspondence problem can also be Automatically Corrected.

(2) These installation problems need to be **Manually Corrected** by the installer according to the error code prompt: (Error codes refer to Table 1)

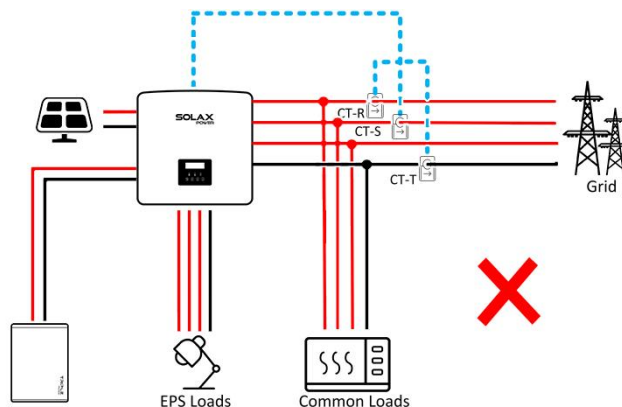
- ① 'CT' has not been successfully clipped on the phase wire
- ② 'CT' is wrongly connected to the loads side



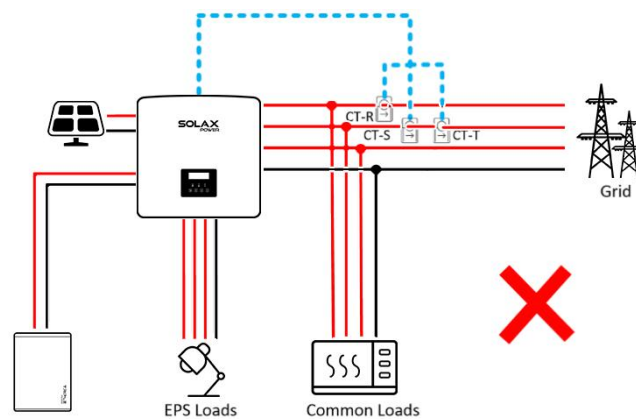
- ③ 'CT' is wrongly connected between the loads and the inverter



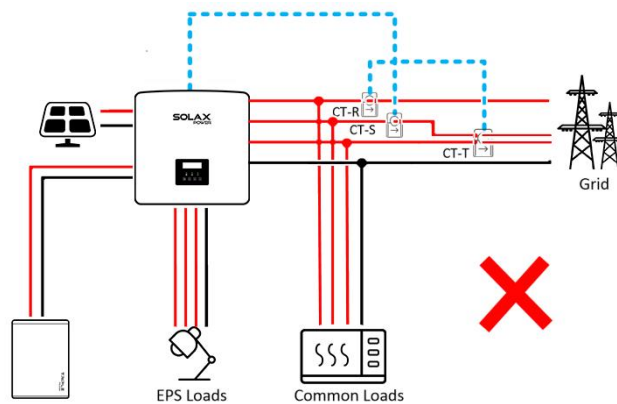
- ④ The CT is connected to the N wire



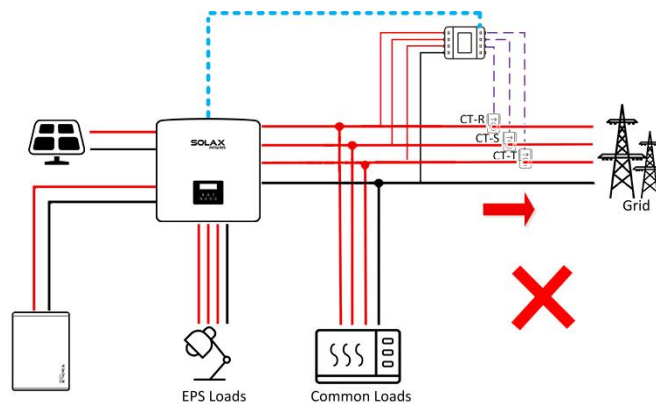
⑤ Two CTs are connected to the same phase.



⑥ The CT is connected to 2 wires

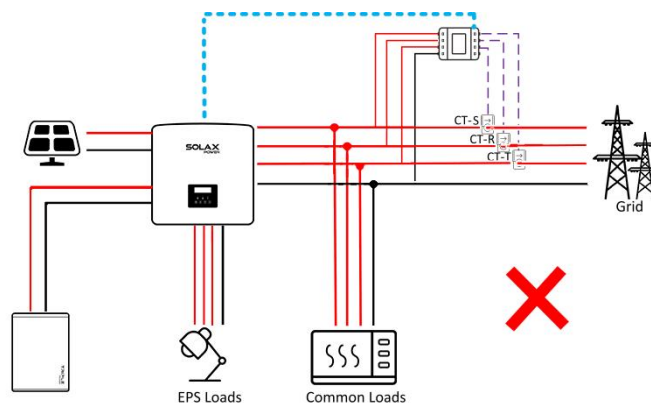


⑦ The direction of 'Meter-CT' is reversed



Note: Meter's direction reverse problem also need to be Manually Corrected according to the error code prompt.

⑧ 'Meter-CT' three phases correspondence is not correct



Note: If there has communication problem with Meter and Meter-CT, the 'Meter fault' will be reported.

How to Use the Meter and CT Self-Check Function?

After the installation is completed, the installer can enable the '**Installation Check**' on the LCD screen or the SolaX Cloud, the inverter will detect whether the Meter and CT has been installed correctly. The inverter will automatically correct it or display the corresponding error code if there has problem.

The Table 1 describes the meaning of error codes.

Tab.1 The Error Codes and Corresponding Meaning

Error Codes	Meaning
Fail 00	Phase A has no CT clipped on it.
Fail 01	Phase A has 2 CTs clipped on it, or there is CT clipped on the N wire.
Fail 02	CT1 is clipped to Phase A and Phase B.
Fail 03	CT2 is clipped to Phase A and Phase B.
Fail 04	CT3 is clipped to Phase A and Phase B.
Fail 05	Phase B has no CT clipped on it.
Fail 06	Phase B has 2 CTs clipped on it, or there is CT clipped on the N wire.
Fail 07	CT1 is clipped to Phase A and Phase C.
Fail 08	CT2 is clipped to Phase A, CT1 is clipped on Phase B and Phase C.
Fail 09	CT3 is clipped to Phase A, CT1 is clipped on Phase B and Phase C
Fail 10	CT2 is clipped to Phase A and Phase C.
Fail 11	CT1 is clipped to Phase A, CT2 is clipped on Phase B and Phase C
Fail 12	CT3 is clipped to Phase A, CT2 is clipped on Phase B and Phase C
Fail 13	CT3 is clipped to Phase A and Phase C.
Fail 14	CT1 is clipped to Phase A, CT3 is clipped on Phase B and Phase C
Fail 15	CT2 is clipped to Phase A, CT3 is clipped on Phase B and Phase C
Fail 16	Phase C has no CT clipped on it.

Fail 17	Phase C has 2 CTs clipped on it, or there is CT clipped on the N wire.
Fail 18	No reactive power was detected on phase A after adjusting the CT sequence.
Fail 19	No reactive power was detected on phase B after adjusting the CT sequence.
Fail 20	No reactive power was detected on phase C after adjusting the CT sequence.
Fail 21	Meter phase A direction is reversed.
Fail 22	Meter phase B direction is reversed.
Fail 23	Meter phase C direction is reversed.
Fail 24~31	Reservation
Fail 32	Not connected to the grid
Fail 33	The inverter reports an error and cannot run (see Error for details)
Fail 34	The inverter can not start normally (detection timeout)
Fail 35	ARM and DSP have not established effective communication.
Fail 36	CT is not connected to L phase

Note: The error message is only a reminder and does not necessarily correspond to the actual installation problem. If the problem is not found after the error is reported, please contact Solax service.

Setting Path

Advanced->Meter/CT Setting->Meter/CT Check->Installation Check

Error Display

① Solax Cloud:

The 'Alarm' page on the Solax Cloud will show the "CT/Meter Check Fault" if there was Meter and CT installation problems. The detailed error code is shown on the **Installation Check** page on the **LCD screen**.

Inverter Analysis Alarm History Battery Analysis Inverter Data Statistic Report								
Inverter: H34A1 Registration No.: SSAV Update Time: 2023-09-25 17:13:36 Monitoring data refresh interval: 5 min								
Report Time: 2023-08-25 2023-09-25 Search Export								
No.	Registration No.	Inverter SN	Username	Error Type	Error Name	Report Time	Recovery Time	Operation
1	SR	H34A1	n	Inverter Error	CT/Meter Check Fault	2023-09-08 08:57:15	2023-09-08 09:01:45	Q
2	SR	H34A1	n	Inverter Error	CT/Meter Check Fault	2023-09-07 19:15:16	2023-09-08 08:56:56	Q
3	SR	H34A1	n	Inverter Error	CT/Meter Check Fault	2023-09-07 19:15:16	2023-09-08 08:56:56	Q
4	SR	H34A1	n	Communication Error	CT/Meter Check Fault	2023-09-07 19:15:16	2023-09-08 08:56:56	Q
5	SR	H34A1	n	Inverter Error	CT/Meter Check Fault	2023-09-07 19:10:51	2023-09-07 19:10:56	Q
6	SR	H34A1	n	Communication Error	CT/Meter Check Fault	2023-09-05 13:33:28	2023-09-05 13:35:35	Q
7	SR	H34A1	r	Communication Error	CT/Meter Check Fault	2023-09-05 13:32:48	2023-09-05 13:33:28	Q
Total 7 10/page < 1 > Go to 1								

② LCD Screen

On the Installation Check page, the 'CT Status' shows the current checking status. There are 4 statuses: **Disable / Checking / Success / Fail xx**, as shown in the figure. (Disable: Installation Checking is not enabled; Checking: Checking in progress; Success: Checking passed; Failed xx: Meter and CT has installation problem required manual adjustment)



The Meter/CT Self-Check function of X1-Hybrid/Fit G4

The Meter/CT Self-Check function of X1-Hybrid/Fit G4 has 2 differences from three-phase system in the following ways:

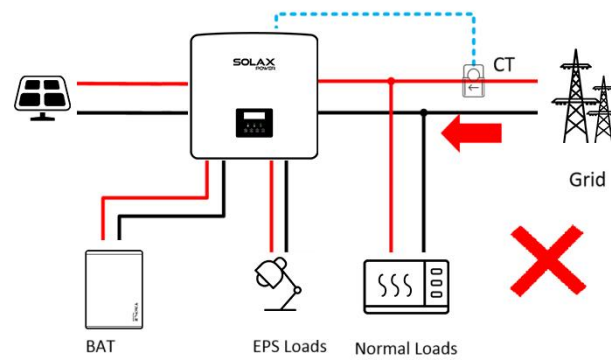
- I. X1-Hybrid/Fit G4 only check CT and **does not check meter**;
- II. Since X1-Hybrid/Fit G4 is single-phase inverter, there is **no phase sequence corresponding issue**.

Here are the installation problems check of the single-phase system.

(1) Automatically Correct

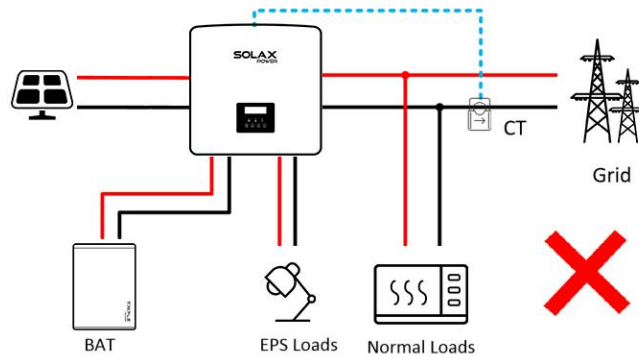
If 'CT direction reversed' is detected, the inverter will **Automatically correct** the CT

direction.

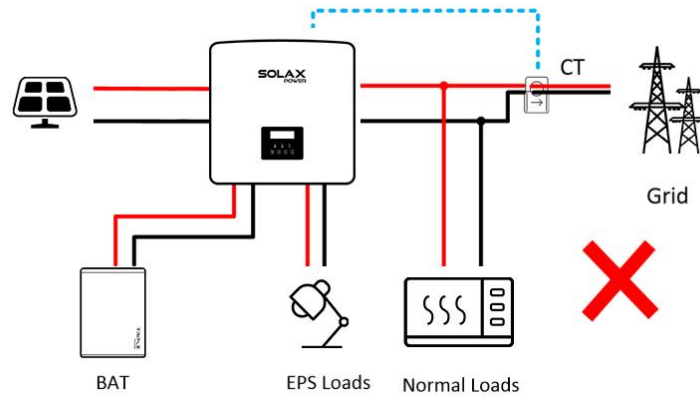


(2) These problems will report '**miss CT fault**' and need to be **Manually Corrected** by the installer.

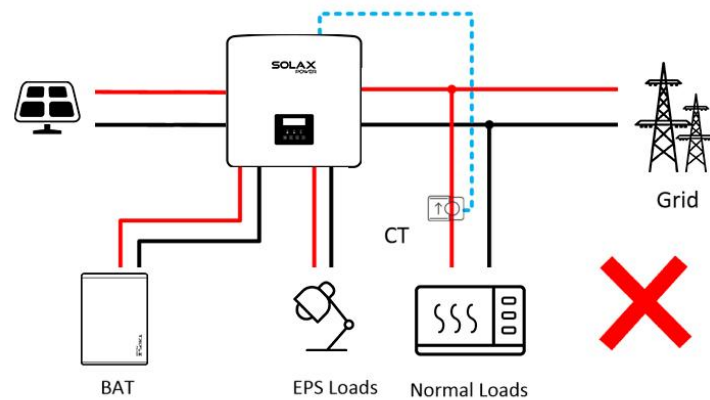
- ① CT has not been successfully clipped on the phase wire
- ② CT is wrongly connected to the N wire



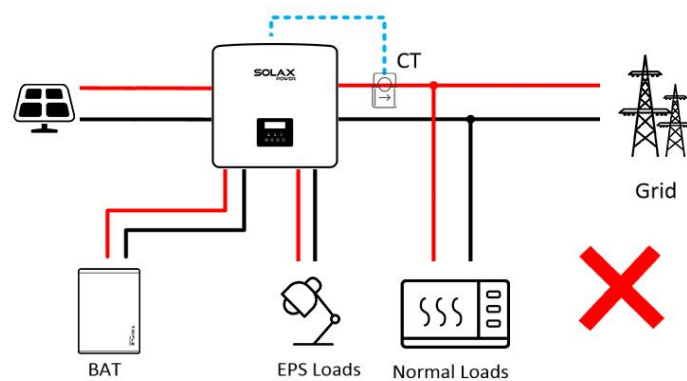
- ③ CT is connected to two wires



- ④ CT is wrongly connected to the loads side



⑤ CT is wrongly connected between the loads and the inverter



Configuration

Compatible Inverter Model:

Function	Model
Meter/CT Self-Check	X1-Hybrid/Fit G4
	X3-Hybrid/Fit G4
	X3-ULTRA
	X1-IES
	X3-IES
	X3-AELIO
	X3-HYB-G4 PRO

Firmware Version:

Model	ARM	DSP
X1-Hybrid/Fit G4	≥1.34	≥1.35
X3-Hybrid/Fit G4	≥1.28	≥1.30
X3-ULTRA	≥018.02	≥017.05
X1-IES	≥009.12	≥013.12
X3-IES	≥012.05	≥016.04
X3-AELIO	≥004.04	≥004.05

X3-HYB-G4 PRO	≥005.11	≥005.12
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Note: Both Inverter Parallel System and AC Coupling (with third-party string inverters) System all support Meter/CT Self-Check function.